ADITYA ITNAL

AEROSPACE ENGINEERING STUDENT

ABOUT ME

Purdue University engineering student with extensive hands-on experience in aerospace systems, CFD (OpenFOAM, Ansys Fluent), and quantum computing research. Skilled in SOLIDWORKS, MATLAB, competitive rocketry, and precision hardware fabrication with proven Eagle Scout leadership.

(972) 876-5256

adityaitnal@gmail.com

Frisco, TX | W. Lafayette, IN

adityaitnal.com

EDUCATION

BS in Aeronautical & Astronautical Engineering

Aug. 2026 - May 2030

- Purdue University
- West Lafayette, IN

SKILLS

- **CFD & FEA:** OpenFOAM, Ansys Fluent, FEA, ParaView, RocketCEA
- **CAD:** Autodesk Fusion 360, SOLIDWORKS, Onshape
- **Programming:** MATLAB, Simulink, Java, Python, Bash
- **Fabrication & Hardware:** Composites, Laser Cutting, 3D Printing, RC Powertrains (ESC, Brushless Motors, Servos)
- **Platforms & Systems:** Linux Server Administration, Git, IBM Quantum, NVIDIA cuQuantum

LEADERSHIP

Eagle Scout, Troop 259

- Led a community service project
- Mentored junior Scouts in leadership and technical skills

Philmont Crew Lead, 2026

- Directed crew navigation, daily logistics, risk mitigation, and team decision-making across 60mi. of rigorous backpacking terrain

EXPERIENCE

Purdue Undergrad Propulsion Lab (PURPL) Sep. 2026 - Present

Igniter Engineer

- Selected propellants and sized the injector geometry, chamber, and nozzle for an igniter used in a larger rocket engine, running the performance calculations in RocketCEA (NASA CEA)
- Designed the igniter in Fusion 360, optimizing for heat and energy output

Purdue Aerial Robotics Team (PART) Sep. 2026 - Present

EDF Engineer

- Designed a 120mm, 11-blade electric ducted fan (EDF) to produce 100 N of thrust at 90 m/s, using MATLAB and Onshape

Anastom May 2026 - Present

Project Engineer

- Engineered a self-healing thermal protection system (TPS) concept using thermodynamic heat flux calculations and finite element analysis (FEA) to evaluate re-entry thermal loads
- Modeled parametric 3D CAD assemblies in SOLIDWORKS, optimizing component geometry for structural integrity under high-temperature gradients

Research on Quantum Computing Platforms for CFD Jul. 2025 - Feb. 2026

Researcher under Dr. Kursat Kara, OSU

- Investigated quantum acceleration platforms (IBM Quantum & NVIDIA) for accelerating CFD solvers, evaluating hybrid QPU-GPU architectures to solve non-linear PDEs (Navier-Stokes)
- Analyzed quantum linear solvers, qubit scaling limits, and error-free GPU emulation in a 7-page technical evaluation report for academic and aerospace research viability

The American Rocketry Challenge (ARC) Jun. 2024 - Jul. 2026

Universe and Beyond

- Designed, fabricated, and launched 3 competition rockets using Kevlar, epoxy, and laser-cut balsa fins profiled to airfoil shapes
- Assembled AeroTech RMS 24/60 rocket motors for 21 team launches, double-checked to ensure launch safety
- Developed a MATLAB trajectory simulation modeling aerodynamic drag, atmospheric density, and wind shear to predict target apogee